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Selection of representative calibration sample sets for near-infrared reflectance spectroscopy to predict nitrogen concentration in grasses

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ABSTRACT

The effect of using representative calibration sets with fewer samples was explored and discussed. The data set consisted of near-infrared reflectance (NIR) spectra of grass samples. The grass samples were taken from different years covering a wide range of species and cultivars. Partial least squares regression (PLSR), a chemometric method, has been applied on NIR spectroscopy data for the determination of the nitrogen (N) concentration in these grass samples. The sample selection method based on NIR spectral data proposed by Puchwein and the CADEX (computer aided design of experiments) algorithm were used and compared. Both Puchwein and CADEX methods provide a calibration set equally distributed in space, and both methods require a minimum prior of knowledge. The samples were also selected randomly using complete random, cultivar random (year fixed), year random (cultivar fixed) and interaction (cultivar × year fixed) random procedures to see the influence of different factors on sample selection. Puchwein's method performed best with lowest RMSEP followed by CADEX, interaction random, year random, cultivar random and complete random. Out of 118 samples of the complete calibration set, 19 samples were selected as minimal number of representative samples. RMSEP values obtained for subsets selected using Puchwein, CADEX and using full calibration set were 0.099% N, 0.109% N and 0.092% N respectively. The result indicated that the selection of representative calibration samples can effectively enhance the cost-effectiveness of NIR spectral analysis by reducing the number of analyzed samples in the calibration set by more than 80%, which substantially reduces the effort of laboratory analyses with no significant loss in prediction accuracy.
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1. Introduction

In the analysis of near infrared spectra, a multivariate calibration model represents a mathematical expression that relates component concentrations to the absorbances of a set of known reference samples at more than one wavelength or frequency [1]. To obtain adequate calibration models, good calibration data and efficient estimation procedures are required [2]. The calibration data should cover the concentration range of the future samples to be analyzed. Usually, the higher the number of samples in the calibration set, the better the prediction ability [3] is because a large number of samples provide more information about the sources of variability that are expected to be present in new samples than a lower number of samples. However, the cost and time required for the reference analysis are often limiting factors when building a calibration set. It is therefore of interest to be able to optimize this by finding a good selection of samples which should be included in reference analysis. Zhou et al. [4] have shown that in some situations the use of representative samples in multivariate calibration improves the model compared to the all sample model. Selection of representative calibration samples is important for the success of the statistical modeling, and it is important that the samples selected for the calibration set cover the variation of future samples. A good calibration set is decisive for accurate chemical determinations in future samples.

When many samples and their corresponding spectra are available, a representative calibration set covering the calibration signal space (meaningful information) can be built. Several approaches are available for selecting representative calibration samples for experimental design and regression validation have been published since the early 1970s. One algorithm that can be used for the selection is based on the D-optimal concept [5,6]. The D-optimal procedure minimizes the variance of the regression coefficients. This is similar to maximizing the covariance matrix, this means that samples are selected in such a way that the variance is maximized and the correlation is minimized. However, as this requires reference data, the practical use of this algorithm is limited. One of the other methods is the CADEX (computer aided design of experiments) algorithm, a sequential method that should cover the experimental region uniformly and is meant for use in experimental design [7]. The CADEX algorithm selects the sample which is closest to

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the mean, and the next sample selected will be the one most distant from the already selected sample. The distance used in the CADEX algorithm is the Euclidean distance. Snee [8] proposed the DUPLEX method, a modification of the CADEX algorithm. The objective of the DUPLEX algorithm is to select representative calibration and test sets that cover the regions with approximately similar statistical properties. This method starts with selecting the two points which are farthest away from each other and are assigned to the calibration set. From the remaining points, the two objects which are farthest away from each other are included in the test set. Alternation between calibration and validation continues until all the samples have been assigned.

Næs [9] proposed another procedure based on cluster analysis. The clustering is continued until the number of clusters matches the number of calibration samples desired. In another method proposed by Puchwein [10], samples are first sorted according to the Mahalanobis distance to the center of the set. A limiting distance is defined between samples. The samples that are closer to the selected point than the limiting distance are removed. The most extreme sample among the remaining samples is selected and this procedure is continued until there are no data points left. Among the other methods proposed is unique-sample selection via NIR spectral subtraction [11]. It is based on the use of Gauss–Jordan curve fitting to remove common spectral features, and at each step the largest residual feature identifies the sample with the most different spectrum. Another way to reduce the number of samples for a calibration is the SELECT algorithm, which is an approach similar to Puchwein's method employed by WinISI software [12].

A selection procedure that selects samples more or less equally distributed over the calibration space will lead to a flat distribution. For an equal number of samples, such a distribution is more favorable from a regression point of view than the normal distribution, so that the loss of predictive quality may be less than expected when looking only at the reduction of the number of samples [13].

Among these proposed methodologies, the CADEX, DUPLEX, SELECT and Puchwein's methods need a minimum of prior knowledge. In addition, they provide a calibration set homogeneously distributed in space (flat distribution). In the present study, the method proposed by Puchwein and the CADEX algorithm are used. The Puchwein method not only selects samples, but an unambiguous mathematical criterion is also reported to arrive at an optimal subset size and to predict/classify any future unknown samples. The method iteratively eliminates the samples from a large data set. The SELECT and DUPLLEX algorithms were not used in this study, as SELECT approach is similar to Puchwein method and DUPLEX is a modification of the CADEX algorithm developed with the objective to divide the data set into calibration and validation sets that cover the factor space in a similar fashion. In the present study, a unique independent test set is used to validate all the methods as this will give a significant comparison between the different selection procedures.

PLSR results obtained using subsets selected by the Puchwein and CADEX sample selection procedures were compared with results using full calibration set. These results were also compared to the results from subset selection using four different random selection methods with and without constraints. The aim is to study the effects of using reduced representative calibration sets on the development of near-infrared reflectance (NIR) spectroscopy models for the prediction of nitrogen (N) concentrations in grasses.

2. Materials and methods

2.1. Materials

The data set consisted of 1020 grass straw samples taken at harvest of the grass seeds from field experiments carried out at Research Centre Flakkebjerg, Denmark, in 2006, 2007 and 2008. These samples were taken from a wide range of second-year grass species and cultivars, namely smooth-stalked meadow grass (*Poa pratensis* L. cv. Evora), red fescue (*Festuca rubra* L. cv. Maxima), orchard grass (*Dactylis glomarata* L. cv. Amba), perennial ryegrass (*Lolium perenne* L. cv. Pimpernel and Calibra).

The samples were dried for 48 h at 80 °C before grinding using a Cyclotec1093 Sample mill (FOSS, Hillerød, Denmark). Thereafter an aliquot of the straw powder (5 mg) was used for chemical analysis of N using a combustion method (Vario EL III, Elementar, Germany). Results are given as percent total N of dry matter.

2.2. NIR measurements

The samples were measured by NIR spectroscopy (Q-interline Spectroscopic Analytical Solutions, QFAflex 600; Tølløse, Denmark) in reflectance mode. Log (1/R) was recorded within the wavelength range from 1100 nm to 2498 nm (4000 cm^{-1} to 9091 cm^{-1}). Spectra were obtained using 128 scans at 16 cm^{-1} resolution.

2.3. Multivariate data analysis

Principal component analysis (PCA) was performed to explore the data for outliers and clustering of the samples according to species, cultivars and years. The 1020 samples included 4 replicates of all treatments in each experiment. The treatments were different N application rates and N application strategies. Chemical analysis was performed for 50 samples, which were selected using stratified random sampling from each replicate giving $50 \times 4 = 200$ samples, considering years and cultivars as stratum within each replicate.

In order to select a good set of samples to form the basis of the calibration set, several methods were tested in order of complexity:

a. Random selection, without any constraints (complete random method).
b. Constrained random selection making sure that each year is present an equal amount of times (cultivar random method).
c. Constrained random selection making sure that each cultivar is present an equal amount of times (year random method).
d. Constrained random selection making sure that both the years and the cultivars are equally present in the calibration set (interaction random method).
e. The method proposed by Puchwein.
f. CADEX algorithm.

2.4. Puchwein's algorithm

The Puchwein procedure begins with data reduction of the original NIR data matrix by PCA. Thus, instead of representing the data matrix by a $200 \times K$ matrix, the matrix size is reduced to $200 \times A$, where K represents the number of variables (wavelengths) of the original data and A is the optimal number of PCA components chosen. The data were not pre-processed prior to sample selection this is in order to include diverse samples of both physical and chemical variation, and also according to scatter.

The Mahalonobis distance of each sample to the center of gravity for the data matrix is calculated, and the samples are sorted according to their distance from the center. In addition, the distance between each sample is calculated. The algorithm then goes through the following steps:

1. Define a limiting distance between samples.
2. Find the most extreme sample having the largest distance to the center.
3. Set all the samples which are within the limiting distance away from the sample selected in step 2 as redundant samples.
4. Go back to step 2 and find the next sample on the list of extreme samples.
5. When the end of the data set is reached, go back to step 1 and increase the limiting distance.

The number of selected samples depends on the size of the limiting distance: if it is small, many samples will be included; if it is large, very few samples will be included. The procedure can therefore be repeated several times for different limiting distances until the limiting distance is reached for which the desired number of samples is selected.

According to Puchwein a good starting value for the limiting distance (D_1) is

$$D_1 = k.(p-2)$$

$$D_m = m.D_1$$

Where k is a constant (Puchwein set this to 0.2), p is the number of principal components, and m denotes the cycle number (i.e. how many times the above steps 1 through 5 are repeated).

Since the number of selected samples depends on the size of the limiting distance, different values for k were tested, ranging from 0.02 and up to 2.5. It was found that $k = 0.05$ gave desirable results as the number of samples eliminated in each cycle was meaningful.

In order to decide where to stop the sample reduction, two plots are used. The first plot shows the number of samples that are removed in each cycle and the total number of samples left. The second plot shows the theoretical sum of leverages (each sample has the same leverage) together with the true sum of leverages and the difference between these.

2.5. CADEX algorithm

The Kennard and Stone [7] algorithm sequentially selects a subset of samples that are uniformly distributed over the factor space based on the Euclidean distance between objects. Let k be the optimal components chosen in the PCA model, n be the number of samples and **T** be the score matrix of dimension n x k.

The squared Euclidean distance (D^2) from the ith to the jth object is defined as

$$D_{ij}^2 = \|t_i - t_j\|^2 = \sum_{q=1}^{k} \left(t_{qi} - t_{qj}\right)^2 \quad (1)$$

The object which is nearest to the mean of the $\mathbf{T}_{n \times k}$ can be considered as the most representative of this given data set, it is included as the first point in the calibration set C of size m and denoted as P_1. Then among the remaining samples, the second object for the training set is the one situated farthest away from P_1. The third object chosen is the one that is farthest away from both P_1 and P_2, etc.

Let $P_1, P_2, \ldots, P_w$ ($w<m$) be w samples that have been assigned to C. The next object P_{w+1} added to this set is the object from the remaining (m–w) samples that is farthest away from the samples already added to C using the following expression:

$$\Delta_{w+1}^2 = \max_{i \neq j} \left\{\Delta_i^2(w)\right\}, \text{for } i = w+1, \ldots, m \quad (2)$$

Where

$$\Delta_i^2(w) = \min_r \left\{ D_{1i}^2, D_{2i}^2, \ldots, D_{qi}^2 \right\} i \neq r$$

Since there is no definite mathematical criterion in the CADEX algorithm to get an optimal subset size, the subsets of size $m = 100$, 80, 60, 40, 19 and 15 corresponding to 85%, 68%, 51%, 34%, 16% and 13% of the total number of samples, respectively were generated.

Once the minimal number of samples (m) was decided using Puchwein's procedure, m samples were randomly selected using 20 iterations in four different situations, namely complete random, cultivar random, year random and interaction random as described above in order to see the effect of Puchwein and CADEX sample selection algorithms.

To test how representative the selected samples are, 40% of the samples were randomly chosen in order to validate the developed model using the selected subsets according to Puchwein, CADEX, random selection procedures and the model using full calibration data. Root mean square error of cross-validation (RMSECV) plotted against partial least squares regression (PLSR) components using different pre-processing methods was used to select the optimum pre-processing method and optimum number of PLS components in the PLSR model. Pre-processing methods included Savitzky–Golay first derivative (1d) and second derivative (2d) [14] averaging over 7 and 9 points, respectively, using a second order polynomial, multiplicative signal correction (MSC) [15] and extended multiplicative signal correction (EMSC) [16].

3. Results

Raw NIR spectra of 1020 samples showed large variations and clear peaks (Fig. 1a). PCA score plots showed no clear groupings according to species, cultivars or years (data not shown). Raw spectra of the 200 samples selected for chemical analysis are shown in Fig. 1b after the removal of two obvious outliers. The 198 remaining samples were used in the sample selection procedure. The original data matrix was transformed into a new reduced one with 10 PCs, explaining 99.99% of the total variation. Fig. 2 show the number of samples left over and eliminated in each cycle using Puchwein's procedure. In the first cycle, no samples were eliminated, in the second step one sample was eliminated, 8 to 24 samples were eliminated during the steps 3 to 9, while only none to 4 samples were removed from step

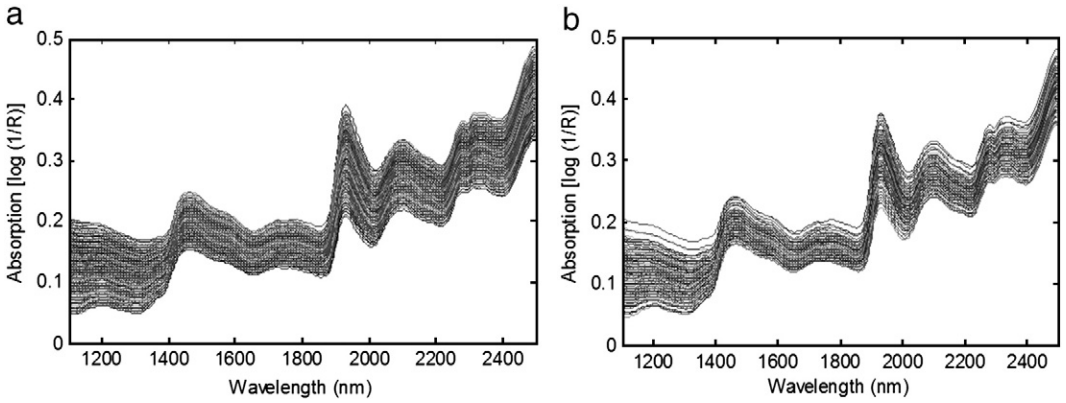


Fig. 1. Raw spectra of all samples ($n = 1020$) (a) and raw spectra of selected samples for chemical measurements ($n = 198$) after removing 2 outliers (b).

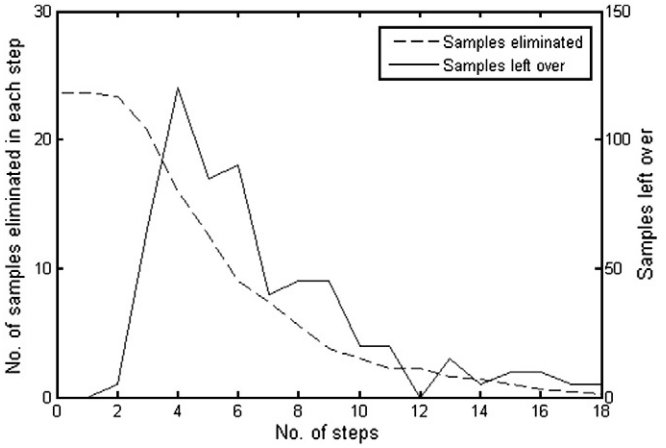


Fig. 2. Samples left over and samples eliminated in each step against number of steps.

10 and onwards. This indicates that the remaining samples after step 9 are already far apart and that the number of samples left is rather low. Fig. 3 shows the theoretical sum of leverages, observed sum of leverages for samples left over and difference between these two plotted against the number of samples eliminated. The curve of differences between theoretical and observed sum of leverages of left-over samples increases drastically up to 19 samples left (step 9), and thereafter it starts decreasing slowly until 11 samples are left over for a few cycles and finally drops off sharply as very influential samples are removed in the end. The number of samples left over at the stage of decrement corresponds well to the minimal samples observed in Fig. 2. Therefore, 19 samples were chosen as the minimal representative samples.

The results of the PLSR models obtained using the complete calibration set (step = 0) and reduced subsets for steps 3 to 11 and for step 13 using Puchwein's procedure are shown in Table 1a. The original 118 samples can be reduced to 19 with no significant loss in the accuracy of prediction. The predicted % N plotted against measured % N for the complete set and reduced subset of 19 samples showed a clear linear correlation (Fig. 4). In Table 1b, the PLSR model results generated using CADEX algorithms for different pre-defined subsets are shown along with the full calibration set. Prior to building the calibration, data were pre-processed using the MSC pre-processing method. MSC was applied separately for each subset used for the CADEX algorithm and subset selected during different numbers of cycles for Puchwein's procedure. Both selection procedures resulted in models of similar

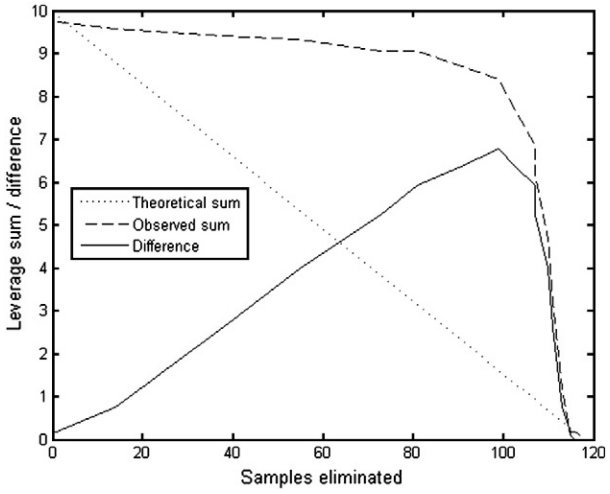


Fig. 3. Theoretical sum of leverages, observed sum of leverages for samples left over and difference between these two plotted against number of samples eliminated.

Table 1a
PLSR results predicted for percent nitrogen (%N) by using original data set (step = 0) and for reduced subsets for step 3 to 11 and for step 13 using Puchwein's sample selection procedure.

Step	Range (% *N*)	Mean (%N)	SD (%N)	Samples left over	# PLSR comp.	R^2	RMSEP
0	0.31–1.59	0.82	0.25	118	6	0.872	0.092
3	0.32–1.59	0.84	0.25	104	6	0.867	0.093
4	0.39–1.59	0.83	0.25	80	6	0.857	0.099
5	0.41–1.59	0.84	0.24	63	5	0.860	0.099
6	0.41–1.59	0.85	0.26	45	5	0.864	0.095
7	0.41–1.59	0.83	0.26	37	5	0.860	0.096
8	0.49–1.59	0.85	0.27	28	5	0.869	0.096
9	0.49–1.59	0.82	0.27	19	5	0.865	0.099
10	0.49–1.59	0.82	0.28	15	4	0.876	0.095
11	0.53–1.59	0.88	0.31	11	4	0.812	0.123
13	0.59–1.22	0.83	0.21	8	3	0.605	0.188

performance compared with the model fitted using all samples. However, when more and more samples were removed, Puchwein's procedure resulted in lower RMSEP compared with the CADEX algorithm, which appears from clearly from Tables 1a and 1b, due to decreased variation of % N.

The finally selected 19 samples using Puchwein's procedure represented the variation of the complete calibration set well, including all 5 cultivars (Amba, Pimpernel, Evora, Maxima and Calibra), all 3 years and replicate variation (Tables 2a and 2b) of the complete data set (Table 1a). This was also the case when 19 samples were selected using the CADEX algorithm (Tables 2a and 2c). The range and mean of % N were better covered in the 19 samples selected using Puchwein's method than in samples chosen using the CADEX algorithm (Tables 1a and 1b). The space covered by the selected 19 samples using Puchwein's and CADEX algoriths is shown in PCA score plots in Fig. 5.

In order to assess whether it is important to go through this thorough selection procedure for finding a small representative set of the data, four other schemes were tested as described above (a to d). The results obtained when 19 samples were randomly selected with 20 iterations using these four schemes are shown in Table 3, along with Puchwein's and CADEX algorithms. Puchwein's method performed best with lowest RMSEP followed by CADEX, interaction random, year random, cultivar random and complete random.

4. Discussion

The comparison of the different methods in Table 3 showed approximately similar results for both the Puchwein and the CADEX algorithm, with slightly lower RMSEP for Puchwein's method followed by the random selection methods: interaction random, year random, cultivar random and complete random. The complete random selection is easier to perform, but it is open to the possibility that some source of variation will be lost. Complete random methods are not competitive with all the other methods shown in Table 3. Constrained random methods are performed to see how different factors, for example in the present case: year, cultivar and replicates, influence the sample selection procedure. The order of the random selection method when it comes to model performance was as expected, as the interaction random is the most conservative, while the complete random is the most liberal way of selecting the 19 samples using only the sample information.

The variation of the 19 selected samples using Puchwein's method was similar to that of the 118 samples from which they were selected, while the selected subset exhibited increased standard deviation compared to the original larger set (Table 1a). Although Puchwein's algorithm reduces the number of samples to be used in the training set, it does not reduce the analyte concentration range present in these samples, as can be seen in Table 1a. This implies that

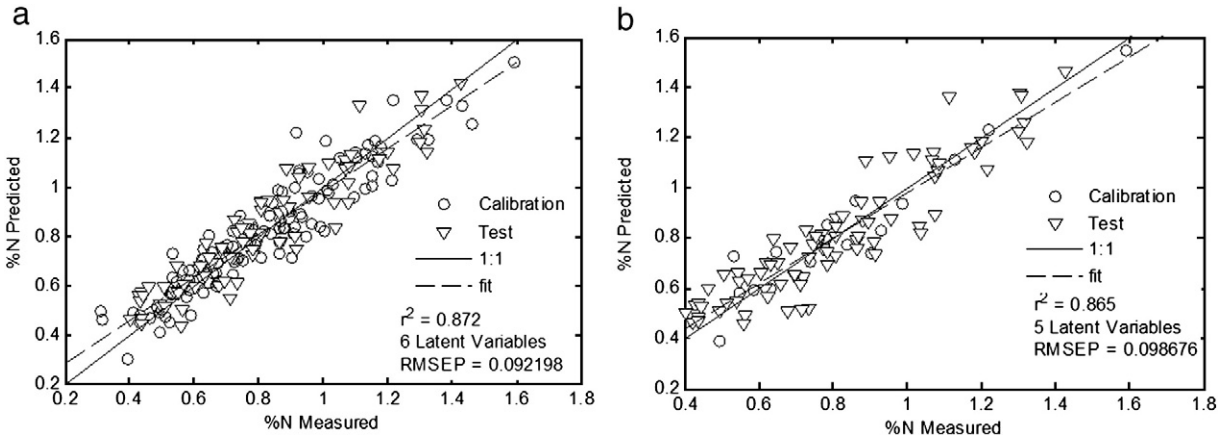


Fig. 4. Predicted vs. measured plot of % N using all calibration samples (a) and using 19 representative samples selected using Puchwein's method (b).

Puchwein's sample selection algorithm selects the unique samples with large variation. These reduced 19 samples also showed all variations of the original set including samples from different year, cultivar, and replicates of experiment (Tables 2a and 2b). The subset chosen using the CADEX algorithm showed a similar variation as the Puchwein method up to the subset was reduced to 34% ($n=40$), after which the variation suddenly dropped (Table 1b). However in the case of Puchwein's procedure the variation decreased steadily, implying that the subset can be reduced further, if necessary. The 19 samples chosen using the CADEX algorithm also included samples from different year, cultivar and replicates of the full calibration set (Tables 2a and 2c).

A better approach to selecting the minimal calibration set so that they are representative of the population is to utilize statistical experimental design or design of experiments techniques in order to ensure that the selected experiments produce the maximum amount of relevant information; however, in most cases only real samples are available, therefore experimental design is not possible. This is the case for the analysis of natural products and for most samples coming from an industrial production process. The labor-effective, inexpensive, ease and speed of NIR technique makes it feasible to screen the large number of samples necessary to ensure a representative training set. Thus, the present sample selection method can be used to replace the more troublesome and time-consuming wet-chemical or labour-intensive sample screening methods or to reduce the effort of expensive and time-consuming chemical analysis.

The manageable size of the calibration set samples will vary according to the cost and time required to analyze each sample by the traditional method. For some fields, such as pharmacy, the preparation and/or analysis of 10 samples may be very costly. In other fields, such as nitrogen analysis, 100 samples can be analyzed with minimal cost. There is no single way to answer the question: what is the smallest sample size in regression, as it depends upon different aspects of the data to be analyzed. Even though sample size is often determined by pragmatic considerations and based upon each individual model, a Monte Carlo simulation performed by Chin and Newsted [17] indicated that PLSR can be performed with a sample size as low as 50. For covariance-based structural equation modeling, it is generally advisable that the "sample size should exceed 100 observations regardless of other data characteristics to avoid problematic solutions and obtain acceptable fit concurrently" [18]. Some researchers even recommend a minimum sample size of 200 cases [19] to avoid results that cannot be interpreted, such as negative variance estimates, i.e. Heywood cases or correlations greater than one, i.e. improper solutions [20]. PLS is applicable even under conditions of very small sample sizes. Isaksson and Næs [21] selected 20

Table 1b
PLSR results predicted for percent nitrogen (% N) by using original data set (step = 0) and for pre-defined reduced subsets using CADEX algorithm.

Calibration set size	Range (% *N*)	Mean (% N)	SD (% N)	# PLSR comp.	R^2	RMSEP
118 (full)	0.31–1.59	0.82	0.25	6	0.872	0.092
100	0.31–1.59	0.83	0.24	6	0.857	0.093
80	0.31–1.59	0.83	0.24	6	0.854	0.098
60	0.31–1.59	0.79	0.23	6	0.853	0.097
40	0.31–1.59	0.79	0.23	5	0.859	0.099
19	0.53–1.16	0.77	0.21	5	0.843	0.109
15	0.53–1.16	0.74	0.21	4	0.824	0.129

Table 2a
Year and cultivar-wise sample details of full calibration set ($n=118$).

Cultivars	Year			
	2006	2007	2008	Total
Maxima	8	8	8	24
Pimpernel	8	8	8	24
Calibra	8	8	9	25
Amba	8	9	8	25
Evora	0	20	0	20
Total	32	53	33	118

Table 2b
Year and cultivar-wise sample details of reduced calibration set ($n=19$) using Puchwein's selection procedure. The numbers in parentheses represent the experimental replicates.

Cultivars	Year			
	2006	2007	2008	Total
Maxima	1 (2)	3 (1, 2, 2)	0	4
Pimpernel	3 (1, 3, 4)	2 (3, 4)	1 (3)	6
Calibra	1 (1)	0	3 (2, 4, 4)	4
Amba	2 (1, 3)	2 (1, 3)	0	4
Evora	0	1 (4)	0	1
Total	7	8	4	19

Table 2c
Year and cultivar-wise sample details of reduced calibration set ($n=19$) using CADEX selection procedure. The numbers in parentheses represent the experimental replicates.

Cultivars	Year			
	2006	2007	2008	Total
Maxima	2 (2, 3)	2 (1, 2)	0	4
Pimpernel	1 (4)	1 (3)	1 (1)	3
Calibra	1 (3)	2 (1, 3)	5 (1, 2, 3, 4, 4)	8
Amba	2 (1,2)	0	1 (1, 4)	3
Evora	0	1 (2)	0	1
Total	6	6	7	19

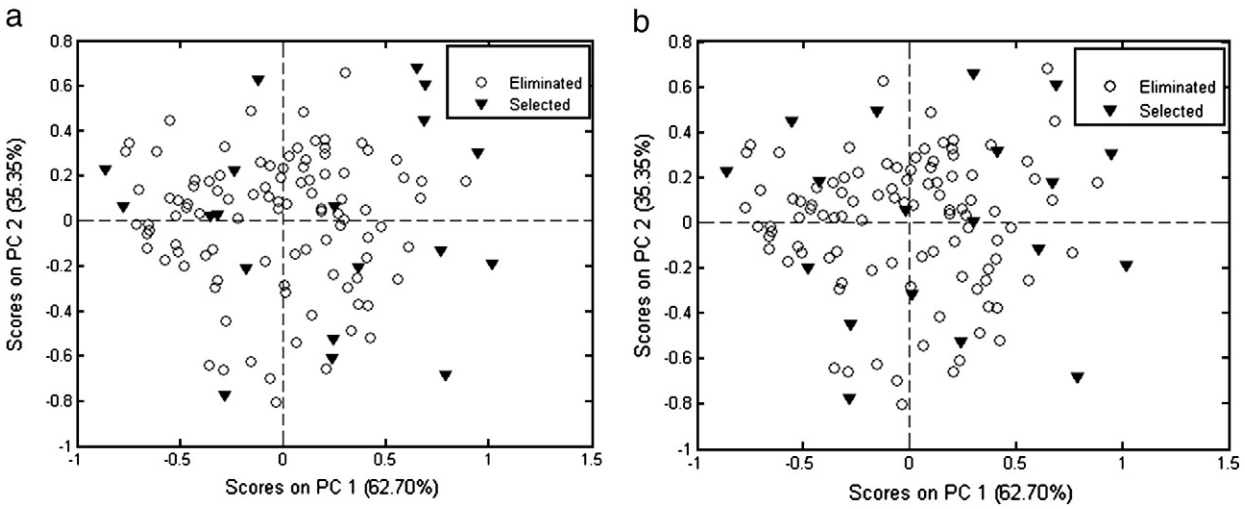


Fig. 5. PCA score plot showing the space covered by selected 19 samples using Puchwein's procedure (a) and using CADEX algorithm (b).

samples as a minimum out of an original calibration set of 114 samples, which gave the same prediction results as the original calibration set based on cluster analysis technique. In the present study, quite similar results were obtained using Puchwein's method, 19 samples were selected as the minimum from the original calibration set of 118 samples. PLSR results obtained using this subset of 19 samples were comparable to the results obtained using 118 samples of the full calibration set, from which these 19 samples were selected. This amount of samples is somewhat smaller than normally used [2]. This indicates that the important aspect of a selection procedure is not the number of samples, but rather how they are selected [19].

PLSR has been shown to be a powerful technique for multivariate calibration of data, when data contain noise, large variable size (even if variable dimensionality is more than the sample size), highly collinear variables and missing observations. However, the PLSR calibration method is strongly influenced by the presence of extreme samples/outliers, and therefore the models obtained may not describe the majority of the data well[22]. Owing to the way in which samples are selected, the performance of both the Puchwein and the CADEX algorithm may be highly influenced by the presence of extreme samples, resulting in overfitting of the model. Hence, before applying these selection algorithms the data set should be inspected for outliers.

Table 3
PLSR results predicted for % N for reduced calibration set of 19 samples using Puchwein's and the CADEX sample selection procedure along with randomly selected samples using different constraints. For random selection methods, PLSR results were obtained for 20 iterations and therefore mean, range and standard deviation (SD) of RMSEP are shown.

Method	Minimal sample size	# PLSR comp range	R^2 range	RMSEP (Mean)	RMSEP (range)	RMSEP (SD)
Puchwein	19	5	0.865	0.099	–	–
CADEX	19	5	0.843	0.109	–	–
Interaction random (Cultivar × Year)	19	3 to 6	0.75–0.89	0.115	0.092–0.142	0.023
Year random (Cultivar fixed)	19	3 to 5	0.72–0.87	0.126	0.101–0.153	0.016
Cultivar random (Year fixed)	19	3 to 6	0.70–0.86	0.131	0.098–0.159	0.014
Complete random	19	3 to 5	0.60–0.84	0.143	0.118–0.195	0.013

A regression model is typically made to do predictions of new/future samples. Therefore, model validation is an essential part of the model development process, if the models are to be accepted and used to support decision-making for future unknown samples. Validating the model will help to estimate the uncertainty of such future predictions. If the uncertainty is reasonably low, the model can be considered valid. Thus, the way in which the validation set is prepared is decisive for making the right conclusion. Samples left over from the selection program are usually used as validation set. The problem is that these are no longer independent samples because the samples left over will be represented in the selected training samples for any good selection algorithm. In the present study, an independent test set is selected from a sufficient number of samples, which are not used in the sample selection procedure to avoid over-optimistic RMSEPs.

5. Conclusions

The results showed that adequate NIR calibrations can be done with quite few samples ($n = 19$) by reducing more than 80% of the original calibration set. A comparison between Puchwein's method and the CADEX algorithm gave approximately similar results with slightly better RMSEP for Puchwein's method. Both of the selection algorithms performed better compared to interaction (year × cultivar) random method, where year-wise and cultivar-wise variation were included. This means that these selection algorithms not only allow a reduction of an initially large set of samples but also select the subset which represents the initial calibration set well.

References

[1] H. Mark, J. Workman, Chemometrics in Spectroscopy, Academic Press, San Diego, Calif, 2007.
[2] T. Næs, T. Isaksson, T. Fearn, T. Davies, A User-Friendly Guide to Multivariate Calibration and Classification, NIR Publications, Chichester, UK, 2002.
[3] A. Lorber, B.R. Kowalski, The effect of interferences and calibration design on accuracy: implications for sensor and sample selection, Journal of Chemometrics 2 (1988) 67–79.
[4] Z.H. Zhou, J.X. Wu, W. Tang, Ensembling neural networks: many could be better than all, Artificial Intelligence 137 (2002) 239–263.
[5] J. Ferré, F.X. Rius, Selection of the best calibration sample subset for multivariate regression, Analytical Chemistry 68 (1996) 1565–1571.
[6] J. Ferré, F.X. Rius, Constructing D-optimal designs from a list of candidate samples, Trends in Analytical Chemistry 16 (1997) 70–73.
[7] R.W. Kennard, L.A. Stone, Computer aided design of experiments, Technometrics 11 (1969) 137–148.
[8] R.D. Snee, Validation of regression models: methods and examples, Technometrics 19 (1977) 415–428.

[9] T. Næs, The design of calibration in NIR reflectance analysis by clustering, Journal of Chemometrics 1 (1987) 121–134.
[10] G. Puchwein, Selection of calibration samples for near-infrared spectrometry by factor analysis of spectra, Analytical Chemistry 60 (1988) 569–573.
[11] D.E. Honigs, G.H. Hieftje, H.L. Mark, T.B. Hirschfeld, Unique-sample selection via near-infrared spectral subtraction, Analytical Chemistry 57 (1985) 2299–2303.
[12] J.S. Shenk, M.O. Westerhaus, Population, definition, sample selection, and calibration procedures for near-infrared reflectance spectroscopy, Crop Science 31 (1991) 469–474.
[13] K.I. Hildrum, T. Isaksson, T. Naes, A. Tandberg, Near infra-red spectroscopy; bridging the gap between data analysis and NIR applications, Ellis Horwood, Chichester, 1992.
[14] A. Savitzky, M.J.E. Golay, Smoothing and differentiation of data by simplified least squares procedures, Analytical Chemistry 36 (1964) 1627–1638.
[15] P. Geladi, D. McDougall, H. Martens, Linearization and scatter-correction for near-infrared reflectance spectra of meat, Applied Spectroscopy 39 (1985) 491–500.
[16] H. Martens, J.P. Nielsen, S.B. Engelsen, Light scattering and light absorbance separated by extended multiplicative signal correction. Application to near-infrared transmission analysis of powder mixtures, Analytical Chemistry 75 (3) (2003) 394–404.
[17] W.W. Chin, P.R. Newsted, Structural equation modelling analysis with small samples using partial least squares, in: R.H. Hoyle (Ed.), Statistical strategies for small sample research, Sage, Thousand Oaks, CA, 1999, pp. 307–341.
[18] F. Nasser, J. Wisenbaker, A Monte Carlo study investigating the impact of item parceling on measures of fit in confirmatory factor analysis, Educational and Psychological Measurement 63 (2003) 729–757.
[19] H.W. Marsh, K.-T. Hau, J.R. Balla, D. Grayson, Is more ever too much? The number of indicators per factor in confirmatory factor analysis, Multivariate Behavioral Research 33 (1998) 181–220.
[20] W.R. Dillon, A. Kumar, N. Mulani, Offending estimates in covariance structure analysis: comments on the causes of and solutions to Heywood cases, Psychological Bulletin 101 (1987) 126–135.
[21] T. Isaksson, T. Næs, Selection of samples for calibration in near-infrared spectroscopy. Part II: selection based on spectral measurements, Applied Spectroscopy 44 (7) (1990) 1152–1158.
[22] I.N. Wakelinc, H.J.H. Macfie, A robust PLS procedure, Journal of Chemometrics 6 (4) (1992) 189–198.